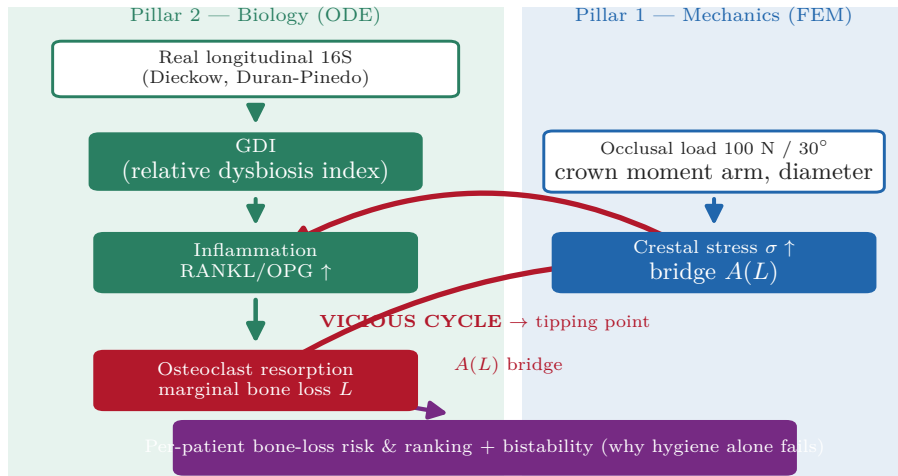


Peri-implantitis as a mechano-biological feedback loop: ecology $\rightarrow$ host $\rightarrow$ mechanics $\rightarrow$ ecology



GDI = guild dysbiosis index (log-ratio of dysbiotic vs. commensal guilds; relative / ordinal). RANKL/OPG = osteoclast accelerator / brake (RANKL: receptor activator of NF- κ B ligand; OPG: osteoprotegerin, a decoy receptor). $A(L)$ = crestal-stress amplification vs. bone loss L – the single bridge FEM $\rightarrow$ ODE. ODE / FEM = disease ODE model / finite-element model.